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Minor Project

1010103494

LinguaAI: Speech-to-Text & AI-Powered Endangered Language Preservation Platform

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Introduction

- Endangered languages face imminent extinction due to declining native speakers, rapid globalization, and a severe lack of accessible digital preservation tools.
- **LinguaAI** is an AI-assisted web platform designed for automated speech-to-text transcription, translation indexing, and digital archiving of endangered and regional languages.
- The platform integrates real-time Web Speech API recognition, custom language database management, analytics dashboards, and an interactive admin control panel.
- Developed under Minor Project Phase I (Course Code: 1010103494, Project ID: M2_CE_045) at Silver Oak University.

Background and Motivation

- Over 3,000 endangered languages worldwide lack digital documentation, leading to irreversible loss of cultural heritage and linguistic diversity.
- Traditional documentation relies on manual field notes and audio recordings, which are slow, inaccessible, and difficult to search or analyze.
- Existing speech-to-text tools primarily target dominant global languages (English, Spanish, Mandarin), completely neglecting minority and regional dialects.
- **Motivation:** Build an accessible, real-time Web-based platform to transcribe, catalog, manage, and preserve endangered languages with administrative oversight and analytics.

Relevance and Importance

- **New Insight:** A unified web architecture combining browser-level Web Speech recognition with MySQL relational indexing for real-time multilingual transcription archiving.
- **Relevant to:** Linguists, cultural researchers, educational institutions, language revival groups, and government cultural departments.
- **Why it matters:** Digital preservation enables real-time acoustic transcription, searchable transcript archives, language status tracking, and secure role-based administrative control.

Literature Review

Sr.No	Ref.No(Citation)	Algorithm used	Dataset	Result	Findings
1	[1] Austin & Sallabank (2011)	Manual Field Recording	Linguistic Field Surveys	Static Audio Archives	Highlights lack of automated web tools
2	[2] Baeovski et al. (2020)	wav2vec 2.0 Neural Net	LibriSpeech Corpus	Low WER (<15%)	Requires heavy GPU; data scarce for dialects
3	[3] W3C Speech API (2020)	Web Speech Engine JS	BCP-47 Language Locales	Zero-latency speech-to-text	Browser execution eliminates GPU server costs
4	[4] M. C. Bird (2011)	Local Language Toolkit	Indigenous Speech Samples	Custom Local Scripts	Lacks web access & relational DB indexing
5	[5] Besacier et al. (2014)	Acoustic/Phonetic Models	Under-resourced Speech	Variable Accuracy	Low-resource datasets limit neural ASR
6	[6] MySQL AB (2023)	Relational Indexing Engine	Multilingual UTF-8 Text	Fast Query Speeds	utf8mb4 enables lossless script storage

Research Gap

- Existing linguistic archives focus primarily on static audio/video storage and lack real-time automated web transcription.
- Commercial ASR engines (Google Cloud, IBM Watson) require expensive API subscriptions and lack support for regional low-resource dialects.
- Heavy offline neural ASR models (e.g., Whisper, Wav2Vec2) demand GPU server hardware inaccessible to field researchers in remote areas.
- Current open-source platforms lack an integrated web dashboard combining live speech capture with endangered language status cataloging (Safe, Vulnerable, Endangered).

Objectives

- Design and implement a responsive web application for real-time speech-to-text transcription.
- Create a centralized MySQL database schema to store users, endangered language metadata, transcriptions, and messages.
- Integrate Web Speech API supporting dynamic BCP-47 language selection and active status toggling.
- Implement role-based access control with secure authentication and admin dashboard management.
- Provide real-time analytics, character/word metrics, and CSV data export capabilities.
- Deploy and test the system across multiple browsers for latency, accuracy, and accessibility.

Required Dataset, Tools and Technology

- **Programming Languages:** JavaScript (ES6+), PHP 8.x, HTML5, CSS3
- **Database Engine:** MySQL / MariaDB (Relational schema with language & transcription indexing)
- **Speech & Web APIs:** Web Speech API (SpeechRecognition engine, BCP-47 language codes)
- **Server & Environment:** XAMPP / Apache Web Server, AJAX / Fetch API
- **Development & Tools:** VS Code, Git & GitHub, Node.js
- **Export & Analytics:** Local XAMPP Web Deployment, CSV Data Exporter, Chart.js System Analytics Engine

Methodology

- **System Architecture:** Client-server architecture with PHP backend processing AJAX requests and interfacing with MySQL database.
- **Speech Processing Pipeline:** Capture microphone input via Web Speech API, perform real-time speech-to-text tokenization, and compute word/character metrics.
- **Database Schema Design:** Define relational tables for users, languages, recognition_languages, transcriptions, and messages with foreign key constraints.
- **Admin Panel & Analytics:** Implement secure admin login, language status management, transcript review, and system usage analytics.
- **Security & Validation:** Enforce input sanitization, password hashing, session isolation, and PDO parameterized SQL queries.
- **Testing & Deployment:** Conduct unit testing (TC-01 to TC-06), AJAX latency testing, and local XAMPP deployment.

Result

- Achieved real-time browser speech-to-text transcription with client-side latency under 120 ms.
- Successfully integrated 49 language locales with active endangerment status filtering (Safe, Vulnerable, Severely Endangered).
- Empirical testing across Chrome, Edge, Brave, and Safari confirmed stable DOM rendering and zero server GPU load.
- User accounts, transcriptions, and contact inquiries are reliably persisted in MySQL database tables.

Conclusion

- LinguaAI successfully provides an accessible, low-resource web platform for real-time endangered language transcription and archiving.
- Minor Project Phase I milestones (55% completed) successfully establish the client-side W3C Web Speech pipeline, MySQL relational data storage, and user authentication modules.
- The browser-native execution model eliminates expensive server infrastructure, making digital preservation practical for researchers and native communities.

Future Work

- **Offline Model Integration:** Embedding WebAssembly (WASM) compiled Wav2Vec 2.0 speech models for zero-connectivity field work.
- **Raw Audio Crowdsourcing:** Extending database schema to store raw audio blobs alongside transcripts for open linguistic datasets.
- **Phonetic IPA Converters:** Adding International Phonetic Alphabet converters for detailed acoustic documentation.
- **Mobile App Deployment:** Packaging the web platform into Progressive Web App (PWA) formats for field tablet deployment.

References

- [1] P. K. Austin and J. Sallabank, *The Cambridge Handbook of Endangered Languages*, Cambridge University Press, 2011.
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